

**INDABAX CAMEROON 2025
BLOOD DONATION CAMPAIGN
HACKATHON PRESENTATION**

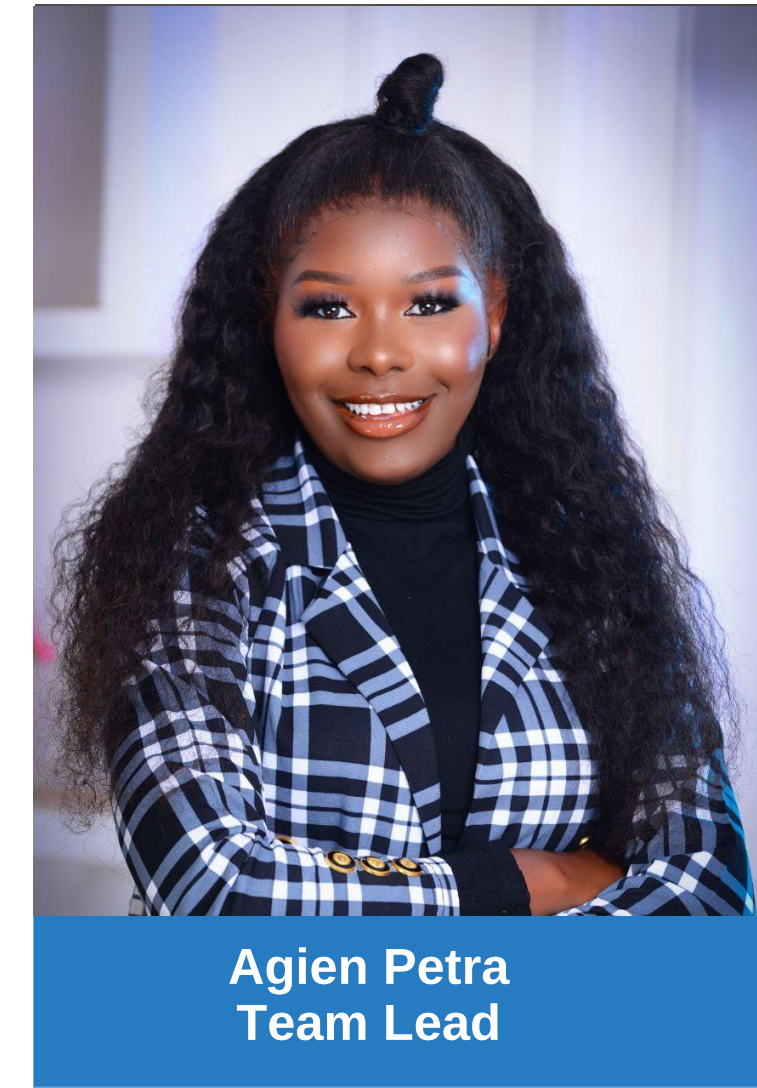
TEAM ABO

Saving Lives through data

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Plan of presentation



Each of us is united by our community engagement, and the need to use Tech for good, especially in the Health Sector.

Every 2 seconds, someone needs blood
Only 38% of the population is eligible to donate
Yet less than 10% actually donate
Campaign managers lack data-driven tools to:
Target potential donors
effectively
Understand donor behavior patterns
Optimize collection initiatives

Problem Statement

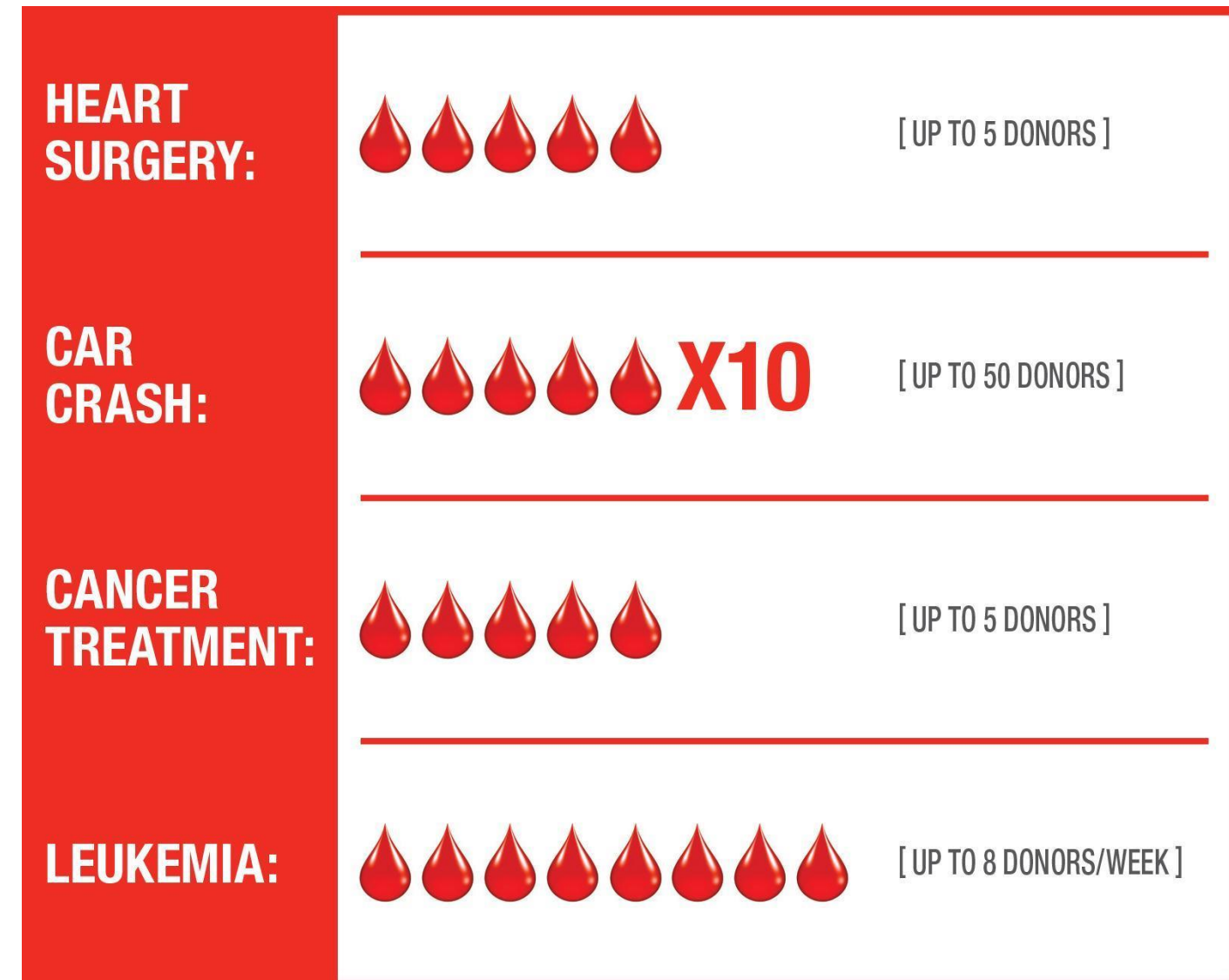
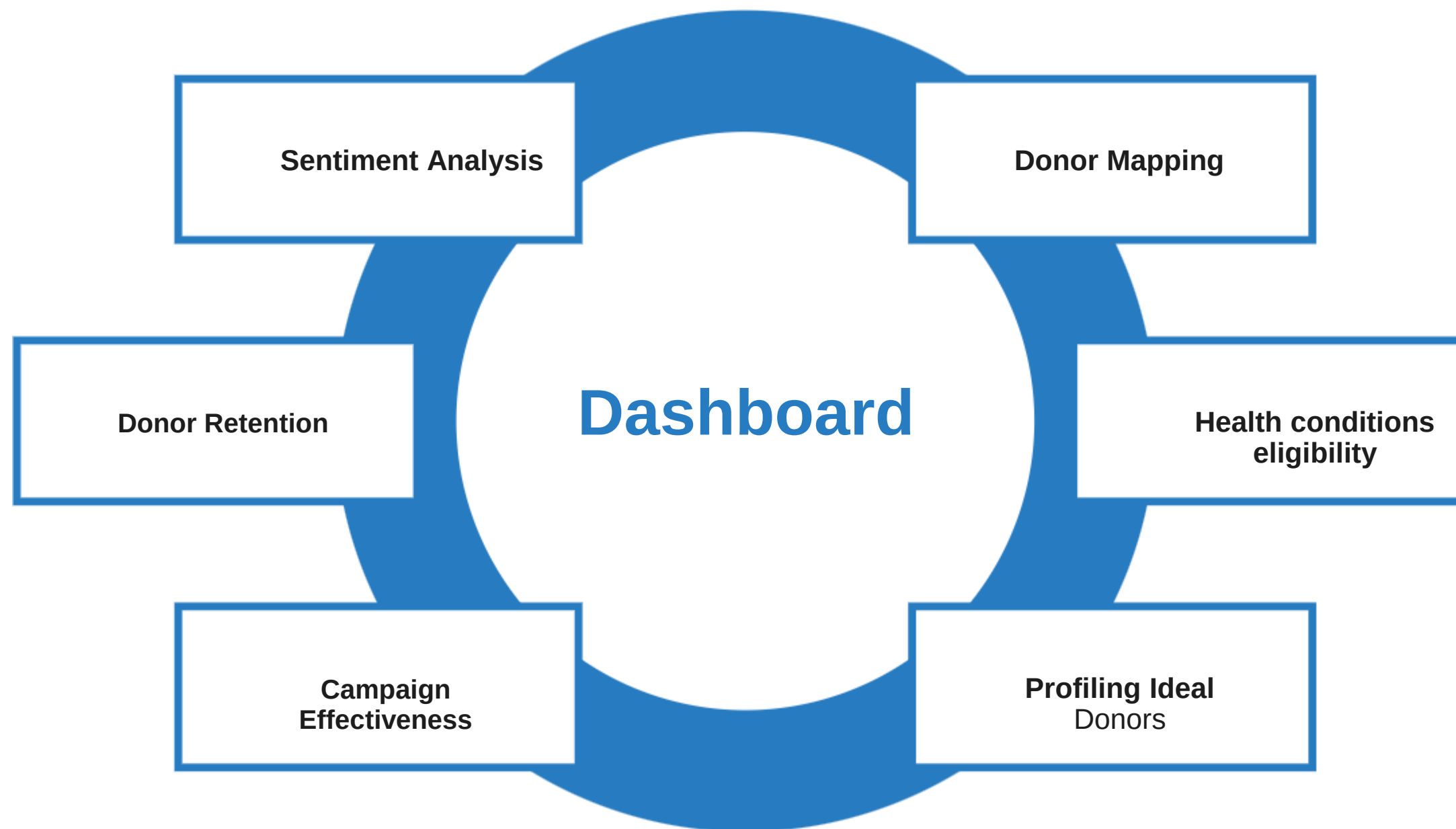


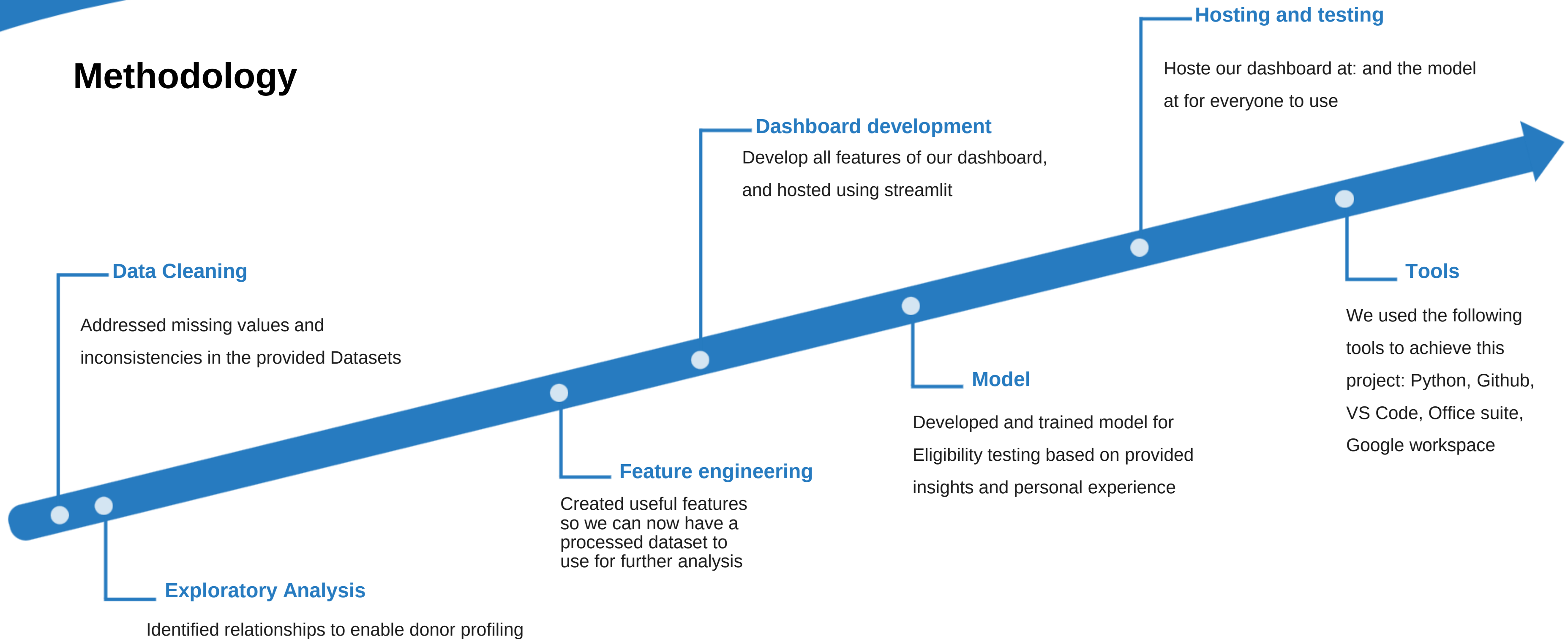
Photo CC: Google Photos

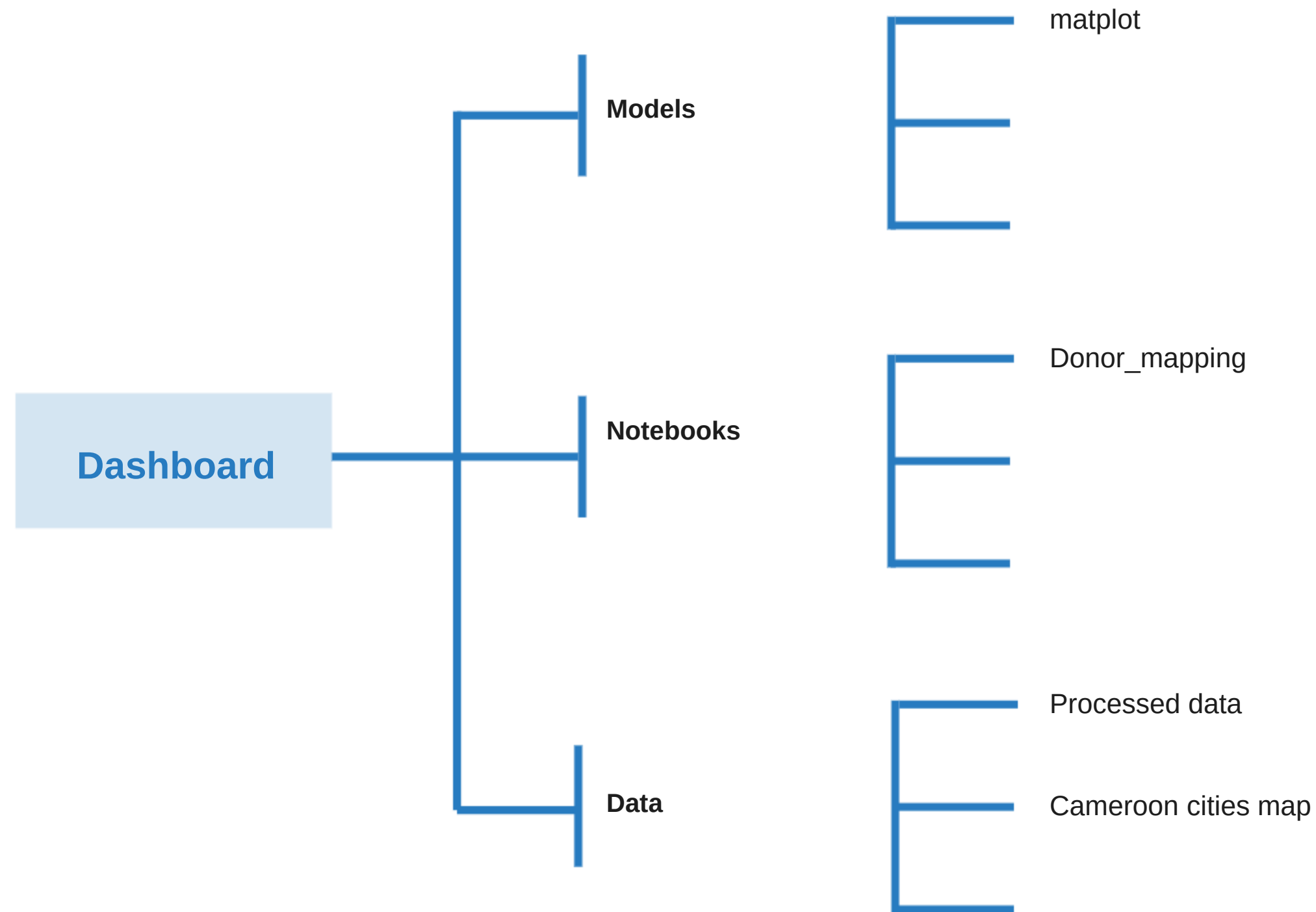
A comprehensive dashboard solution that



- Maps donor distribution geographically
- Analyzes health conditions & eligibility factors
- Profiles ideal donors through clustering
- Measures campaign effectiveness
- Predicts potential donor eligibility

Methodology





Second

Streamlit will open dashboard on your default browser
Visualize the plots and get insights

Launching the Dashboard

Checking Eligibility

Viewing Dashboard

Firstly

Clone the repository
Open VS code, install all packages in requirements.txt
Launch using streamlit run command

lastly

You can check if you are eligible for blood donation on the dashboard.

Software Required



Prerequisites

- **Anaconda:** Python/R distribution for data science, package management.
- **Colab:** Interactive coding environment for data analysis and visualization.
- **Visual Studio Code :** Python IDE for scientific computing and development.
- **Python:** Versatile programming language for diverse applications.
- **Git:** Version control system for tracking code changes.
- **GitHub:** Web platform for Git repositories and collaboration.
- **Streamlit :** Build and share interactive data web apps.
- **Flask:** Containerization platform for application deployment.

Map Donor Distribution

Distribution of Donors Static map

Distribution of Donors interactive map

Health Conditions and Eligibility

Distribution of Donors Static map

Distribution of Donors interactive map

Profiling Ideal Donors

Distribution of Donors Static map

Distribution of Donors interactive map

Campaign Effectiveness

Distribution of Donors Static map

Distribution of Donors interactive map

Donor Retention

Distribution of Donors Static map

Distribution of Donors interactive map

Sentiment Analysis

Distribution of Donors Static map

Distribution of Donors interactive map

To check for eligibility, we hosted the model on And the features take into consideration the Demographics and the health conditions of the potential donor, and return if one is eligible for donation or not.

Key Insights discovered

Geographic targeting

5 arrondissements account for 60% of donations. Douala should be targeted

Age groups

Young professionals (25-35) most underrepresented, though most eligible, so find out why young people shy away from donating

African mythology

Clarifying eligibility myths could increase donor pool by 15%

Support

Blog explaining clear explanations of blood donation and eligibility to donors

Community Engagemnt

Donors unite to save lives through blood donation, forming the foundation of successful campaigns through awareness, participation, and valuable feedback.

Data Science

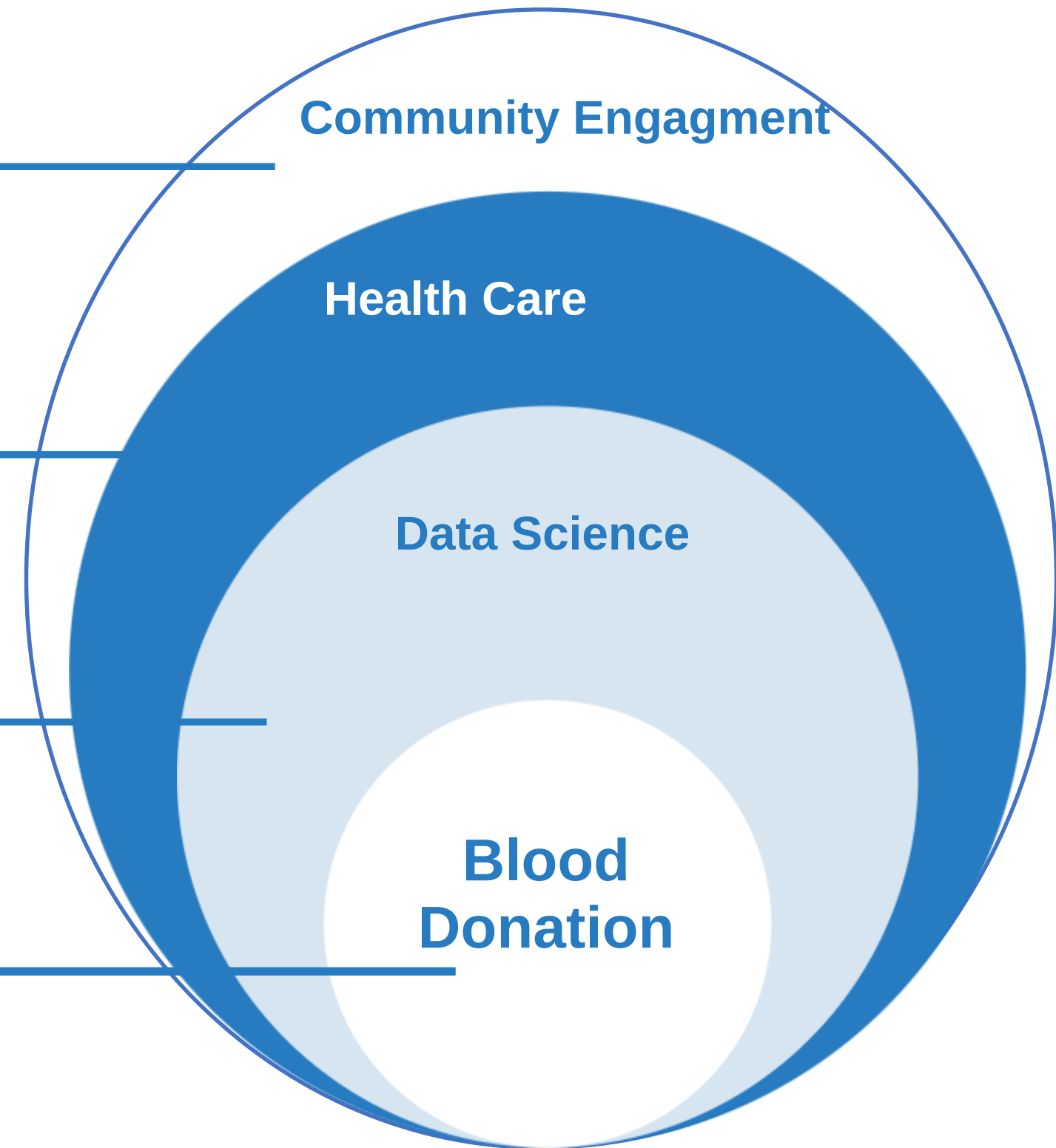
Advanced analytics transforms blood donation data into actionable intelligence, enabling campaign optimization through predictive modeling and pattern recognition.

DEEP LEARNING

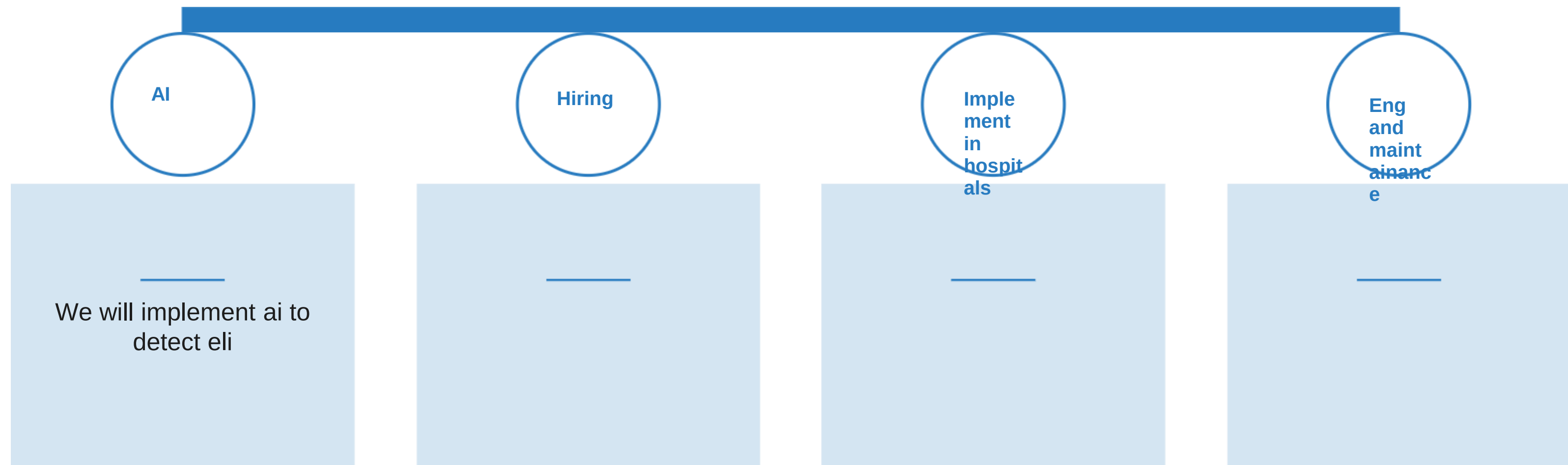
Medical professionals ensure safe blood collection and distribution to meet critical patient needs, with proper screening protocols determining eligibility and allocation.

Blood Donation

Our blood donation filtering allows campaign managers to test scenarios



Future consideration and scalability





Pour le dashboard

Nos Solution

- Solution complète pour l'analyse des campagnes de don de sang
Tableau de bord interactif avec visualisations géographiques
Modèle prédictif pour l'éligibilité des donneurs
Impact potentiel: Augmentation de 30% des dons réussis
Innovation dans l'analyse des sentiments et la segmentation des donneurs

To have a comprehensive overview of our French dashboard, [click here](#).

Any Questions

